

DSA (Abdul Bari) - Day 9

45-day plan | YouTube playlist course

Searching - Linear & Binary

Find an element; binary search needs sorted data.

Key concepts

- Linear search $O(n)$; binary search $O(\log n)$ on sorted data.
- Binary search halves the range each step.
- Careful with mid overflow and boundaries.

Key idea / example

```
mid = lo + (hi-lo)/2; compare and discard half.
```

Today's tasks (1.5h)

- ☐ Implement binary search
- ☐ Trace it on an example
- ☐ Handle not-found

Resources

Playlist: youtube.com/playlist?list=PLsr8vTgyLdy_YndxNcI4WkH5Vorj5qvrV

(Clickable playlist link is in the app: Goals > this day > Study block.)